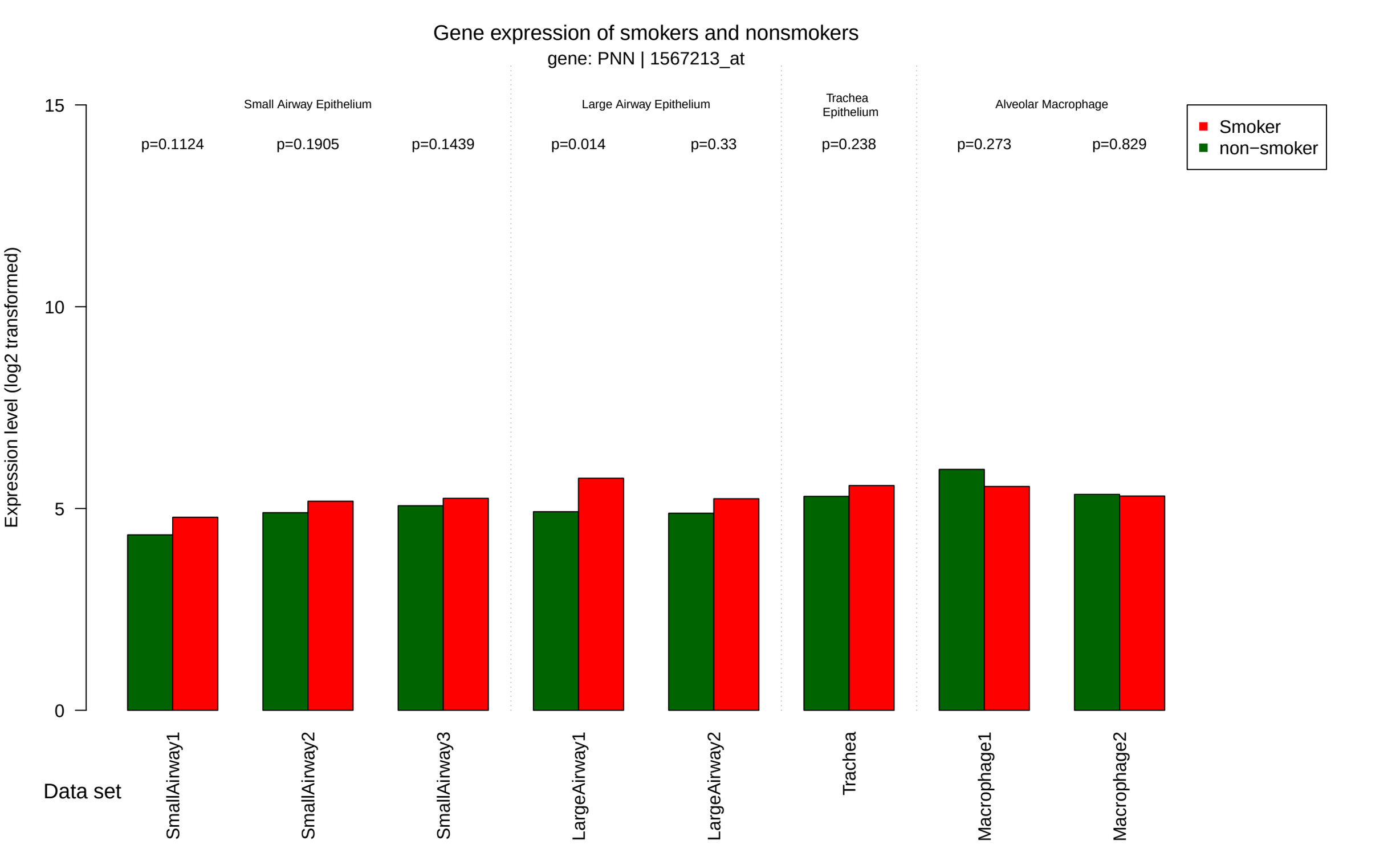
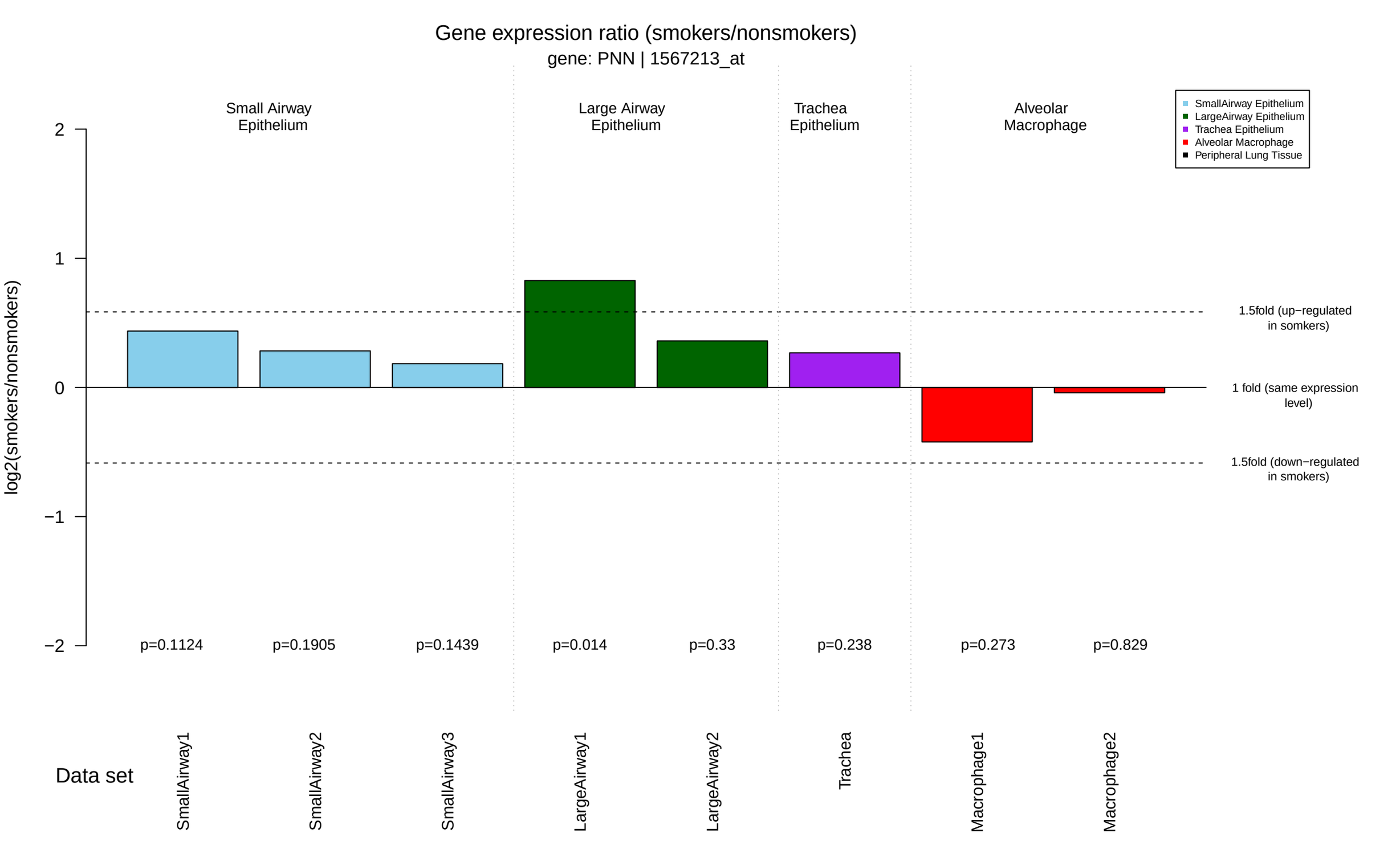
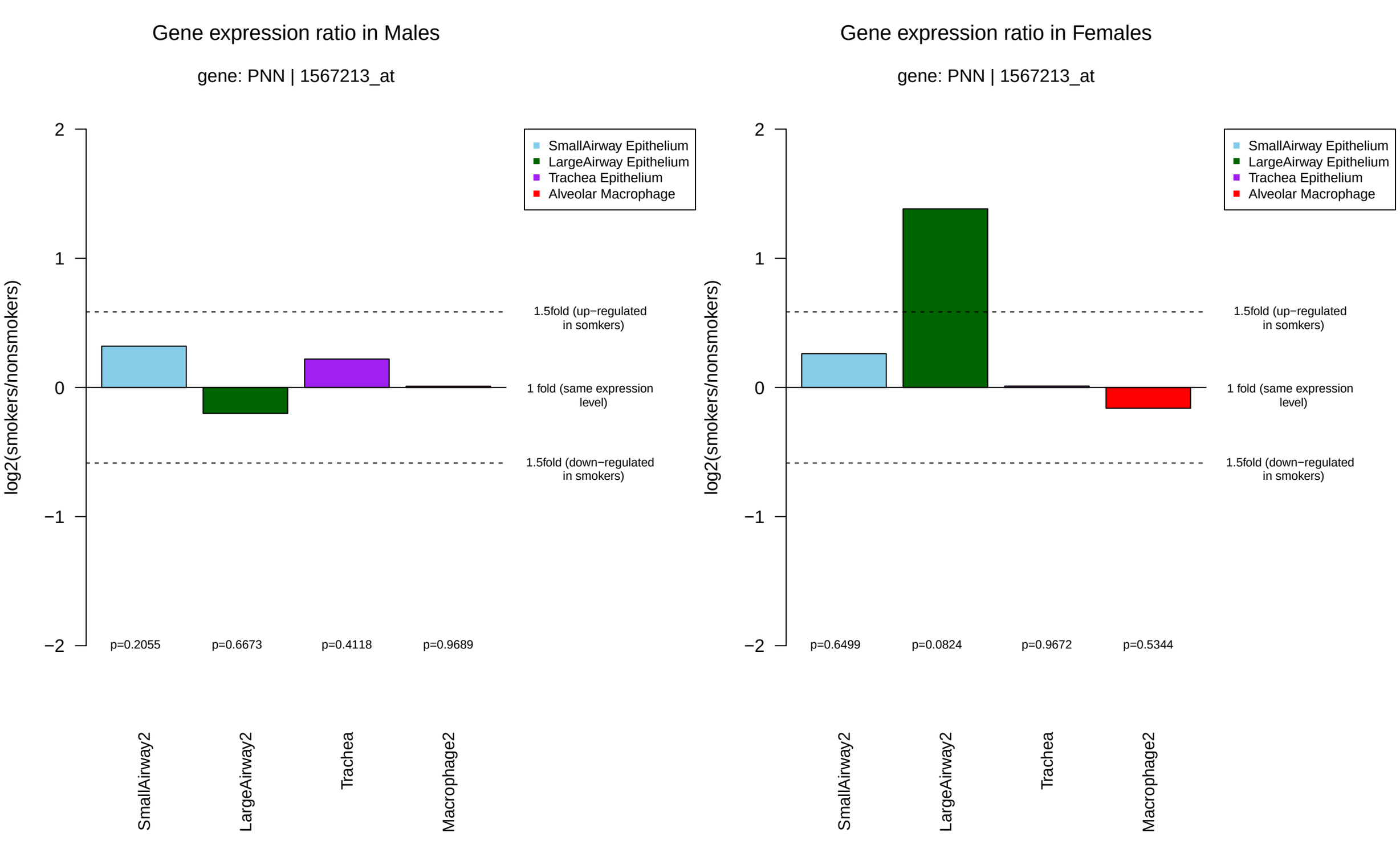
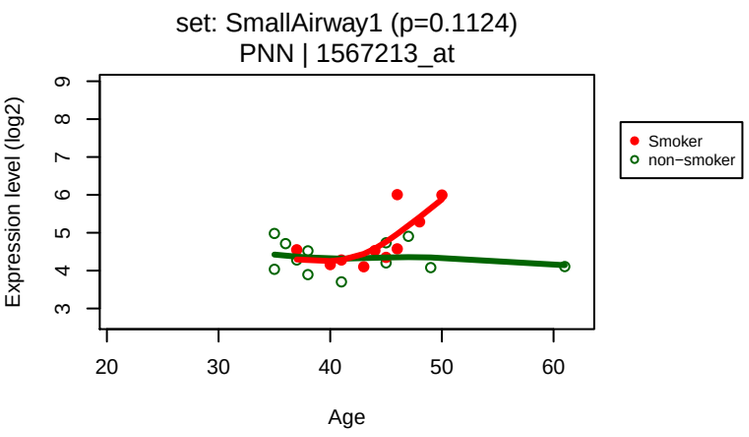
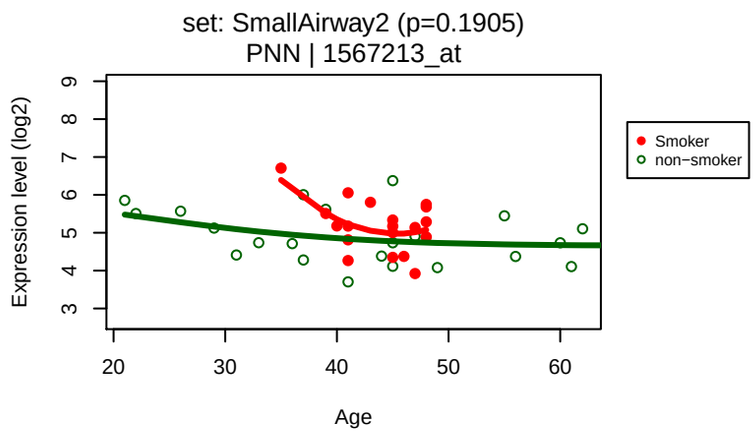
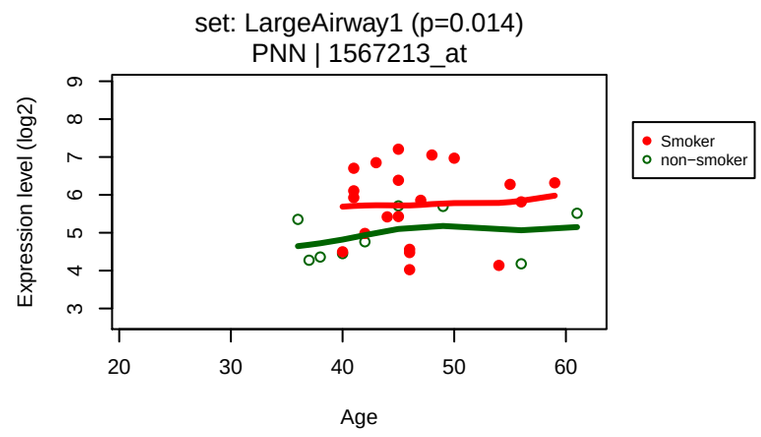

Gene expression of smokers and nonsmokers
gene: PNN | 1567213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1124
p=0.1905
p=0.1439
p=0.014
p=0.33
p=0.238
p=0.273
p=0.829
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

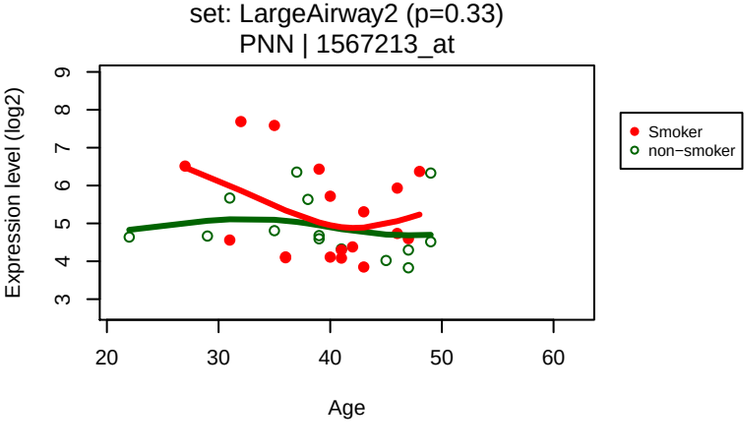

Gene expression ratio (smokers/nonsmokers)
gene: PNN | 1567213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1124
p=0.1905
p=0.1439
p=0.014
p=0.33
p=0.238
p=0.273
p=0.829
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

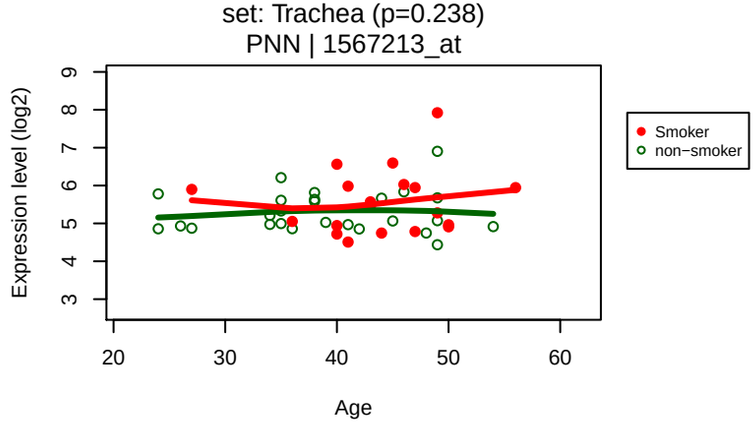

Gene expression ratio in Males
gene: PNN | 1567213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2055
p=0.6673
p=0.4118
p=0.9689
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNN | 1567213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6499
p=0.0824
p=0.9672
p=0.5344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

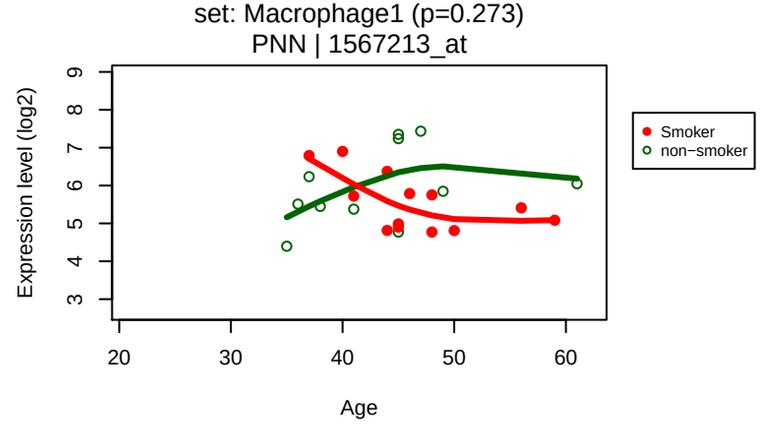

set: SmallAirway1 (p=0.1124)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker

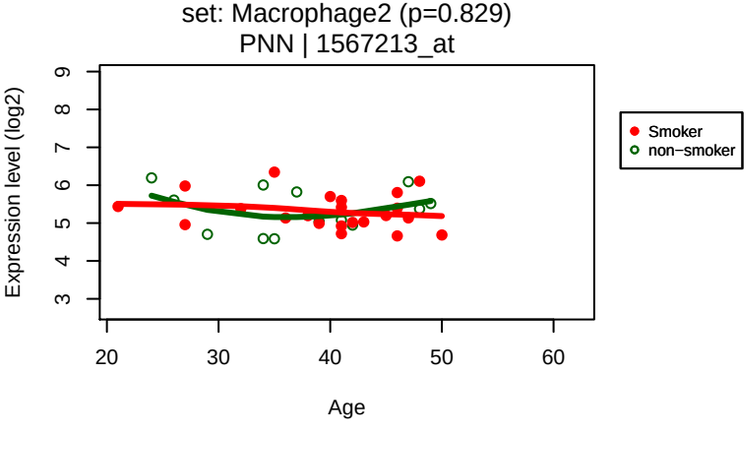

set: SmallAirway2 (p=0.1905)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker

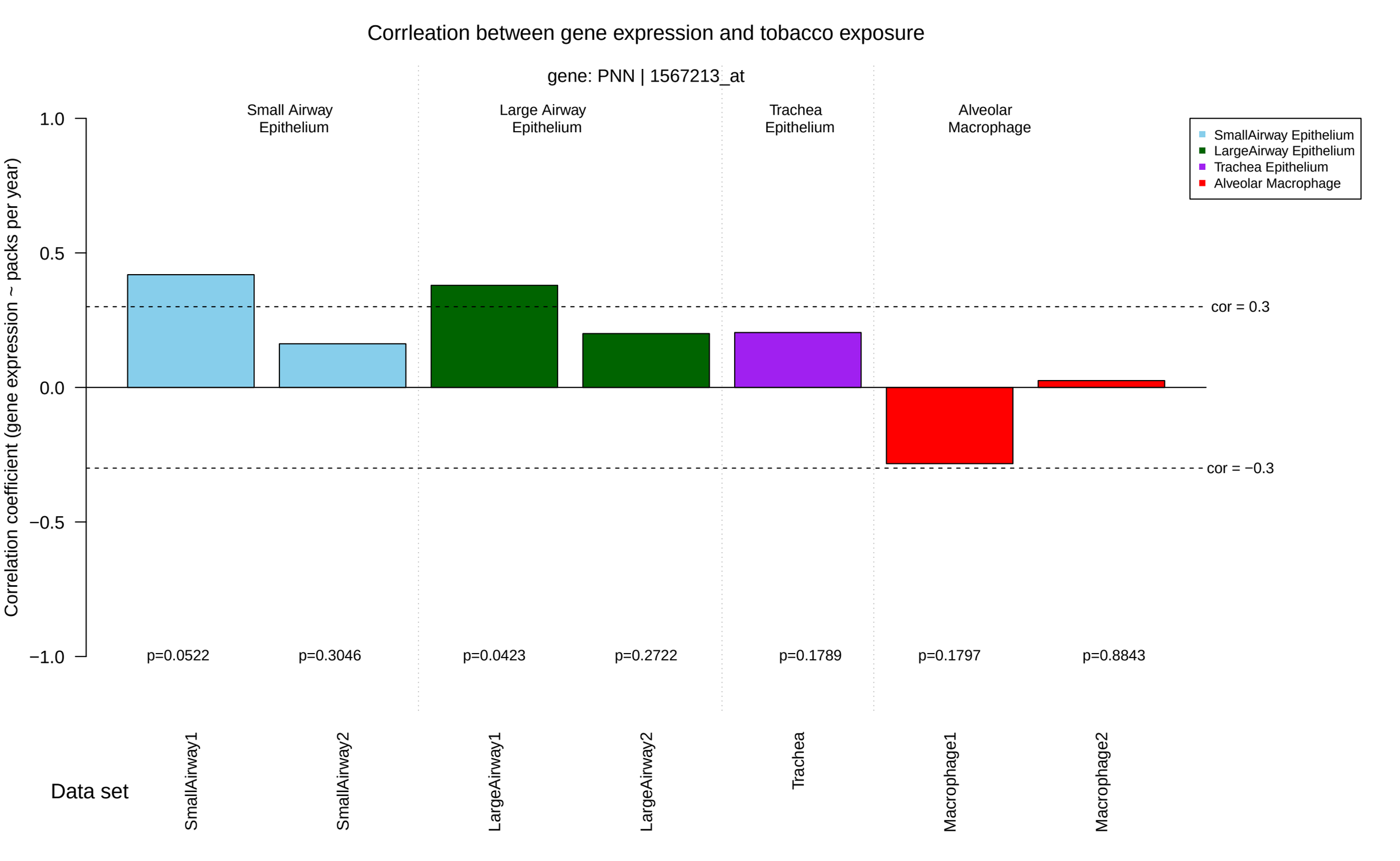

set: LargeAirway1 (p=0.014)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.33)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.238)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.273)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.829)
PNN | 1567213_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNN | 1567213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0522
p=0.3046
p=0.0423
p=0.2722
p=0.1789
p=0.1797
p=0.8843
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2